

Jingeun Kim

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RESEARCH INTERESTS

My research primarily focuses on developing efficient machine learning frameworks. Specifically, I aim to explore efficient models using neural architecture search based on genetic algorithms. Furthermore, I am currently working on model compression, particularly pruning, using genetic algorithms for CNN-based models and pre-trained models.

Keywords: Neural Architecture Search, Genetic Algorithm, Model Compression

EDUCATION

Doctor of Philosophy:

Gachon University, Gyeonggi-do, South Korea (Mar. 2024 -)

Major in IT Convergence Engineering (Advisor: Yourim Yoon)

4.39/4.50 GPA

Master of Science:

Gachon University, Gyeonggi-do, South Korea (Sep. 2021 - Aug. 2023)

Major in IT Convergence Engineering (Advisor: Yourim Yoon)

4.06/4.50 GPA

Thesis title: Machine Learning with Various Feature Selection Methods for Diagnosis of Parkinson's Disease Using RNA-Seq

Bachelor of Science:

Gachon University, Gyeonggi-do, South Korea (Mar. 2016 - Aug. 2021)

Major in Computer Engineering (Advisor: Yourim Yoon)

3.73/4.50 GPA

WORK EXPERIENCES

Gachon University

Sep 2023 – Feb 2024

Master's Degree Researcher

- Optimization of DenseNet architecture using a genetic algorithm with multi-dimensional encoding scheme

PROJECTS

Project 1 (May 2020 - May 2021)

Analysis of Pulmonary Diseases Induced by Fine Particulate Matter and Treatment Drug Screening Using Meta-Heuristic-Based Feature Selection Algorithm

Summary: This study compares how machine learning is efficient when classifying mRNAs or miRNAs.

Role: model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Paper accepted

Project 2 (Mar. 2021 - Nov. 2021)

Development of Next-Generation Battery Management System (BMS) with AI-Based Fire and Explosion Prevention for Lithium-ion Batteries

Summary: Estimate the resistance value of the battery cell after 10 days to prevent explosion and Battery anomaly detection using clustering.

Role: Researcher (model implementation, evaluation)

Advisor: Taegkeun Whangbo

Status: Paper accepted

Project 3 (May 2021 - May 2022)

Theory and Applications of Various Feature Selection Algorithms

Summary: This study uses EEG datasets and genetic algorithms to select meaningful features to increase the accuracy of machine learning.

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Manuscript submitted

Project 4 (May 2022 – July 2024)

Research on IT-BT/ET Convergence Technology Using Meta-Heuristic-Based Feature Selection Algorithms

Summary: Various feature selection methods improve the performance of machine learning when classifying Parkinson's disease

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Paper accepted

Project 5 (May 2023 – July 2025)

Healthway Development for Senior Citizens in the Age of Homo sapiens [Re:]Solution: Building a Healthway for Muscle Atrophy Prevention and Management

Summary: Enhancing classification accuracy through feature selection using correlation analysis

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Manuscript submitted

Project 6 (Jan 2024 - Present)

AI-based Maritime Search and Rescue Decision Making

Summary: This study utilizes AI to improve real-time data analysis, predictive modeling, and resource optimization, enhancing the efficiency and effectiveness of maritime SAR operations.

Role: Overall study design, model implementation, evaluation, interpretation, and manuscript writing

Advisor: Yourim Yoon

Status: Work in progress

Project 6 (Sep 2025 - Present)

Evolutionary Algorithm-Based Lightweight Techniques for Large Language Models

Summary: This project investigates evolutionary algorithm-driven structural optimization

methods for large language models (LLMs) to achieve lightweight architectures while preserving inference accuracy

Role: Principal investigator, Overall study design, experimental evaluation, performance analysis, and manuscript preparation.

Funding: 50 million KRW ($\approx$ USD 38,500) over two years, awarded through the Ph.D. Student Scholarship Program.

Status: Work in progress

PUBLICATIONS

- J6. Pruning for Efficient DenseNet via Surrogate-Model-Assisted Genetic Algorithm considering NAS Proxies
Jingeun Kim, Yourim Yoon
Swarm and Evolutionary Computation
- J5. A Comparison of Binary and Integer Encodings in Genetic Algorithms for the Maximum k-Coverage Problem with Various Genetic Operators
Yoon Choi, **Jingeun Kim**, Yourim Yoon
Biomimetics
- J4. Performance Analysis and Improvement of Machine Learning with Various Feature Selection Methods for EEG-Based Emotion Classification
Sherzod Abdumalikov, **Jingeun Kim**, Yourim Yoon
Applied Sciences
- J3. Improving Modularity Score of Community Detection Using Memetic Algorithms
Dongwon lee, **Jingeun Kim**, Yourim Yoon
AIMS mathematics
- J2. RNA Sequences-Based Diagnosis of Parkinson's Disease Using Various Feature Selection Methods and Machine Learning
Jingeun Kim, Hyejin Park, Yourim Yoon
Applied Sciences
- J1. Comparative Study of Classification Algorithms for Various DNA Microarray Data
Jingeun Kim, Yourim Yoon, Hyejin Park, Yong-hyuk Kim
Genes

CONFERNCES

- C4 Sparse Transformer for Long-term Forecasting via Genetic Algorithm
Jingeun Kim, Yourim Yoon
GECCO 2025 (poster)
- C3. Efficient Pruning of DenseNet via a Surrogate-Model-Assisted Genetic Algorithm
Jingeun Kim, Yourim Yoon
GECCO 2024 (poster)
- C2. Genetic Algorithm-based Pruning for Efficient DenseNet Architecture
Jingeun Kim, Yong-hyuk Kim, Yourim Yoon
ICAIIC 2024
- C1. the battery risk prediction method using the deep learning
Kyunam Choi, Mona O, **Jingeun Kim**, Taegkeun Whangbo
ITC-CSCC 2021

WORKSHOPS

- W1. From Forecast to Action: Uncertainty-Aware UAV Deployment for Ocean Drifter Recovery
Jingeun Kim, Yong-Hyuk Kim, Yourim Yoon
CIKM Workshop on Spatio-Temporal Data Intelligence and Foundation Models 2025 (Oral Presentation)

DOMESTIC PUBLICATIONS

- J2. Enhancing Machine Learning-based Drug-Drug Interaction Prediction using Feature Selection
Hojae Kim, **Jingeun Kim**, Yourim Yoon
Asia-pacific Journal of Convergent Research Interchange
- J1. Battery thermal runaway cell detection using DBSCAN and statistical validation algorithms
Jingeun Kim, Yourim Yoon
The Journal of the Convergence on Culture Technology

MANUSCRIPTS UNDER REVIEW AND IN PREP

- C1. MAC: Memetic Actor-Critic Framework for UAV Path Planning
Jingeun Kim, Seung-ryul Hong, Yong-Hyuk Kim, Yourim Yoon
GECCO 2026
- J1. A Study on the Effectiveness of Feature Selection considering mRMR across Different Datasets
Jingeun Kim, Yourim Yoon
Knowledge and Information Systems

TEACHING EXPERIENCE

2025 Fall	JAVA @Gachone University (T.A.)
2025 Spring	Algorithm @Gachone University (T.A.)
2024 Fall	Discrete mathematics @Gachone University (T.A.)
2024 Spring	Algorithm @Gachon University (T.A.)
2023 Fall	Discrete mathematics @Gachon University (T.A.) Advanced Programming @Gachon University (T.A.)
2023 Spring	Algorithm @Gachon University (T.A.)
2022 Fall	Discrete mathematics @Gachon University (T.A.)
2022 Spring	Algorithm @Gachon University (T.A.)

AWARDS, HONORS & SCHOLARSHIPS

2025-2027	Ph.D. Student Scholarship Program	38,500\$
2025	SCI best paper award	350\$
2024-	Tuition Exemption (100%), Educational Funding Scholarship	4500\$
2021-2023	Tuition Exemption (100%), Educational Funding Scholarship @Gachon University	4500\$
2022	SCI best paper award @Gachon University	100\$
2020	Gachon University x-design contest 3 rd place @Gachon University	200\$
2016-2021	Tuition Exemption (30%), Educational Funding Scholarship	1000\$

@Gachon University

PROFESSIONAL SERVICE

Journal Reviewer	IEEE Transactions on Evolutionary Computation, BMC Psychiatry, Discover Oncology
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SKILLS

Programming Languages	Python, JAVA
AI Frameworks	Pytorch, Tensorflow, sklearn, pandas, numpy
Languages	Korean (Native), English (Intermediate)

REFERENCES

Advisor	Prof. Yourim Yoon yryoon@gachon.ac.kr
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